

■ 2.3.4 Advanced Topic: Patterns for Algebraic Expressions

In trying to set up patterns for algebraic expressions, it is important to understand the standard form that *Mathematica* uses for algebraic expressions.

Here is a typical algebraic expression.

```
In[1]:= -1 / z^2 + 2 (x z)^2 y - z / y
Out[1]= -z^-2 - z/y + 2 x^2 y z^2
```

You can use `FullForm` to see the standard form that *Mathematica* uses for this kind of expression.

```
In[2]:= FullForm[%]
Out[2]//FullForm=
Plus[Times[-1, Power[z, -2]],
Times[-1, Power[y, -1], z],
Times[2, Power[x, 2], y, Power[z, 2]]]
```

The standard internal form that *Mathematica* uses for algebraic expressions is essentially a sum of products of powers.

Any ratio of terms is converted into a product, with denominator terms having negative exponents. When the expression is printed out, however, the terms with negative exponents are again shown in the denominator. In the internal form, however, there are no ratios, as `FullForm` shows.

Patterns that you set up for algebraic expressions must follow the standard form that *Mathematica* uses. There is no point, for example, in having a pattern for a ratio that cannot actually appear in an algebraic expression.

$x_ + y_$	a sum of two or more terms
$n_Integer\ x_$	x multiplied by an integer other than 1
$a_ + b_.\ x_$	a linear expression $a + bx$
$x_ \wedge n_$	x^n with $n \neq 0, 1$
$x_ \wedge n_.$	x^n with $n \neq 0$
$a_ + b_.\ x_ + c_.\ x_ \wedge 2$	a quadratic expression

Some common patterns for algebraic expressions.